AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION 2010

Mathematics Paper II

Time allowed: 2 hours 20 minutes Marks 45

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign that it is correct.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are SEVENTEEN questions. Answer ELEVEN questions in all. Choices are specified inside the paper.

3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a simple calculator if you wish.

(ATTEMPT EITHER Q.1 OR Q.2)

Q.1. (Total 4 Marks)

a. Find TWO rational numbers between 0 and 1 if they exist. Write your answer in fraction form. (2 Marks)

b. Simplify $a^2 \times a^{\frac{3}{4}} \times a^{\frac{1}{4}}$. (2 Marks)

(ATTEMPT EITHER Q.1 OR Q.2)

Q.2. (Total 4 Marks)

If $z = 2 + 3i$ is a complex number, then find:

a. conjugate of $z\left(i.e.\ \bar{z}\right)$ (1 Mark)

b. $z + \bar{z}$ (1 Mark)

c. $z \times \bar{z}$ (2 Marks)

(ATTEMPT EITHER Q.3 OR Q.4)

Q.3. (Total 4 Marks)

a. List the members of the set A, where A = {whole numbers divisible by 3}. (1 Mark)

b. If B = {1,3,5,7} and C = {2,8,10,12}, then find $(B \cup C) - (B \cap C)$. (3 Marks)

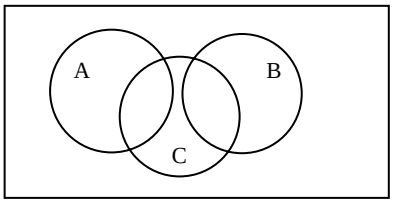
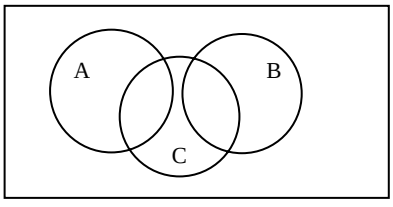
(ATTEMPT EITHER Q.3 OR Q.4)

Q.4. (Total 4 Marks)

Consider the Venn diagrams given in figure i and figure ii. Shade the (i) and (ii) in pencil.

(i) $(A \cup B) \cap C$ (Shade the region $(A \cup B)$ as and $(A \cup B) \cap C$ as)

(ii) $(A \cap C) \cup B$ (Shade the region $(A \cap C)$ as and $(A \cap C) \cup B$ as)

$(A \cup B) \cap C$	$(A \cap C) \cup B$
 Figure i	 Figure ii

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Q.5. (Total 3 Marks)

If $\log(mn) = 2\log m$, then prove that $m = n$.

(ATTEMPT EITHER Q.6 OR Q.7)

Q.6. (Total 4 Marks)

Reduce $\frac{x^2+2x-3}{x^2-3x+2}$ to its lowest form.

(ATTEMPT EITHER Q.6 OR Q.7)

Q.7. (Total 4 Marks)

Prove that $\frac{x+1}{x-1}+\frac{x-1}{x+1}=2+\frac{4}{x^2-1}$.

(ATTEMPT EITHER Q.8 OR Q.9)

Q.8. (Total 5 Marks)

a. If x^4 is divided by $x-3$, then find its remainder. (2 Marks)

b. Factorize completely $16a^4+4b^4$. (3 Marks)

(ATTEMPT EITHER Q.8 OR Q.9)

Q.9. (Total 5 Marks)

a. Factorize completely $6p^2-17pq+10q^2$. (3 Marks)

b. Factorize completely $(c+2)^2-4$. (2 Marks)

(ATTEMPT EITHER Q.10 OR Q.11)

Q.10. (Total 4 Marks)

56 electronic markers are required to assess 1,75,000 scripts in 8 days while working for 5 hours daily. How many e-markers are required for assessing 2,00,000 scripts in 10 days working for 4 hours a day? [assuming rate of script assessment is constant]

(ATTEMPT EITHER Q.10 OR Q.11)

Q.11. (Total 4 Marks)

If $\frac{a}{b} = \frac{c}{d} = \frac{e}{f}$, then prove that $\frac{c}{d} = \sqrt{\frac{a^2 - c^2 + e^2}{b^2 - d^2 + f^2}}$.

Q.12. (Total 5 Marks)

If $A \times B = I$ where $A = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then find matrix *B*.

Q.13. (Total 4 Marks)

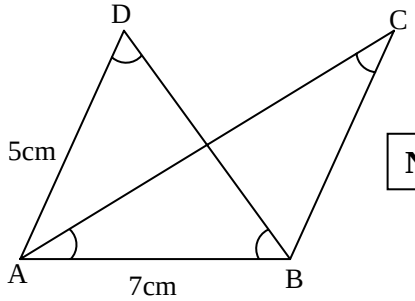
100 students participated in a drawing competition. The participants were marked out of 50. The table below shows information about the participants' marks.

Calculate the arithmetic mean of the average marks obtained by the participants in the competition.

Marks	**Number of Participants**		
1 – 10	9		
11 – 20	13		
21 – 30	15		
31 – 40	36		
41 – 50	27		
Total			

Q.14. (Total 4 Marks)

In the following figure, $\angle BAC \cong \angle ABD$, $\angle ADB \cong \angle ACB$ and perimeter (sum of all sides) of the ΔABC is 20cm. Find $m\overline{BD}$.



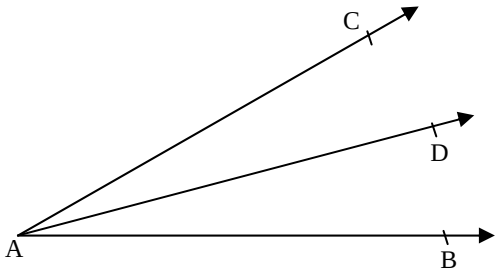
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(ATTEMPT EITHER Q.15 OR Q.16)

Q.15. (Total 4 Marks)

In the following figure, AD is the angle bisector of $\angle BAC$. If P and Q are any two distinct points lying on the angle bisector AD, then write True or False for each statement in the given box.



(i) P and Q are always collinear

(ii) P and Q are always equidistant from any point on AD

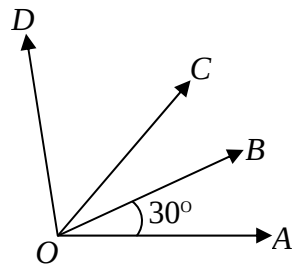
(iii) P and Q are equidistant from AB

(iv) If QB and QC are perpendicular to AB and AC respectively, then $m\overline{QC} = m\overline{QB}$

(ATTEMPT EITHER Q.15 OR Q.16)

Q.16. (Total 4 Marks)

In the following figure, *OB* and *OC* are angle bisectors of $\angle COA$ and $\angle DOA$ respectively. Find the $m\angle DOC$ and $m\angle DOA$.



Q.17. (Total 4 Marks)

Construct a triangle *ABC* with $mAB = 6\text{cm}$, $m\angle A = 60°$ and $m\angle B = 70°$. Also verify that medians of the given triangle are concurrent.

END OF PAPER

Please use this page for rough work